

1. Lecture 1 included a slide that shows “U.S. trading partners 2024”. The source data for this slide is “Exhibit 14. U.S. Trade in Goods by Selected Countries and Areas: 2024” from the PDF included with this assignment. For this problem, create a similar chart but for the top 5 products imported and exported by the US. For the purpose of this problem, define a product as a detailed end-use commodity (there are about 140, see this link for details). You will use an LLM of your choice (ChatGTP, Claude, Gemini, etc) as well as your own non-artificial intelligence to answer this question.
 - (a) Give the LLM a detailed prompt telling it exactly what you want, but not what data source to use. Make any subsequent refinements needed in chat with the LLM. Save the resulting chart.
 - (b) Give the LLM the correct source data, which is “Exhibit 7. U.S. Exports of Goods by End-Use Category and Commodity” and “Exhibit 8. U.S. Imports of Goods by End-Use Category and Commodity” from the included PDF. Chat with the LLM to create the desired bar chart, and look carefully at the results to make sure it is correct. When you’re confident the chart is correct, save it.
 - (c) Compare the charts from part (a) and part (b). Comment on the differences if any.
2. See problem 1 at the end of Chapter 1 of the Feenstra-Taylor book. The navigation instructions for the WTO website are somewhat out of date, but you will be able to answer all the questions starting at the WTO data portal. Answer questions (a) through (i) for the most recent year for which the WTO website has complete data. You will probably find it helpful to use an LLM, along with your brain, to complete this assignment.

If you use an LLM, include a brief description of how you did so. Importantly, be sure to tell the LLM to ONLY use data from the WTO website, and ask it to list the precise link it used to give you your answers. Include the relevant links in your answers to the problems, and verify that the answers given by the LLM match up with what’s on the linked page(s).